4-1

(a)

$N=20 \quad \Sigma Y_i^2 = 7825.94 \quad , \bar{Y} = 19.21 \quad , SSR = 375.47$

$$SST = \Sigma(Y_i - \bar{Y})^2 = \Sigma Y_i^2 - N \cdot \bar{Y}^2 = 7825.94 - 20 \times 19.21^2$$

$$= 445.458$$

$$R^2 = \frac{SSR}{SST} = \frac{375.47}{445.458} = 0.8429 = 84.29\%$$

(b)

$R^2 = 0.7911 \quad , SST = 725.94 \quad , N=20$

$$SSE = SST(1-R^2) = 725.94 \times (1-0.7911) = 151.6489$$

$$\hat{\sigma}^2 = \frac{SSE}{N-2} = \frac{151.6489}{20-2} = 8.4249$$

(c)

$\Sigma(Y_i - \bar{Y})^2 = 631.63 = SST \quad , \Sigma \hat{e}_i^2 = \Sigma(Y_i - \hat{Y})^2 = 182.85 = SSE$

$$R^2 = \frac{SSR}{SST} = \frac{SST - SSE}{SST} = \frac{631.63 - 182.85}{631.63} = 0.7105 = 71.05\%$$

4.3

(a)

$\hat{y}_i = 1.2 + 0.8 x_i \quad , x_0 = 4 \quad \Rightarrow \hat{y}_0 = 1.2 + 0.8 \times 4 = 4.4$ #

(b)

$\hat{\sigma}^2 = \frac{\sum \hat{e}_i^2}{N-2} = \frac{3.6}{3} = 1.2 \quad , \sum (x_i - \bar{x})^2 = 10 \quad , \sum (y_i - \bar{y})^2 = 10$

$Var(f) = \hat{\sigma}^2 \left[1 + \frac{1}{N} + \frac{(x_0 - \bar{x})^2}{\sum (x_i - \bar{x})^2} \right] \overset{x_0 = 4}{=\!=} 1.2 \left[1 + \frac{1}{5} + \frac{(4-1)^2}{10} \right] = 2.52 = Var(\hat{y}_0)$

$se(f) = \sqrt{Var(f)} = \sqrt{2.52} \approx 1.5875$ #

(c)

$\hat{y}_0 \sim N(y_0 , Var(\hat{y}))$

test statistic

$T = \frac{\hat{y}_0 - y_0}{se(\hat{y})} \sim t(3)$

y_0 之 95% 預測區間 : $\left[\hat{y}_0 - se(\hat{y}) \cdot t_{0.025}(3) , \hat{y}_0 + se(\hat{y}) \cdot t_{0.025}(3) \right]$

$= \left[4.4 - 1.5875 \times 3.182 , 4.4 + 1.5875 \times 3.182 \right]$

$= \left[-0.6514 , 9.4514 \right]$

(d)

y_0 之 99% 預測區間 : $\left[\hat{y}_0 - se(\hat{y}) \cdot t_{0.005}(3) , \hat{y}_0 + se(\hat{y}) \cdot t_{0.005}(3) \right]$

$= \left[4.4 - 1.5875 \times 5.841 , 4.4 + 1.5875 \times 5.841 \right]$

$= \left[-4.8726 , 13.6726 \right]$

(e) When $x_0 = \bar{x} = 2$, $\hat{y}_{\bar{x}} = 1.2 + 0.8 \times 1 = 2$

$$Var(f_{\bar{x}}) = \hat{\sigma}^2\left[1 + \frac{1}{N} + \frac{(\bar{x}-\bar{x})^2}{\sum(x_i-\bar{x})^2}\right] = 1.2\left[1+\frac{1}{5}+\frac{0}{10}\right] = 1.44 = Var(\hat{y}_{\bar{x}})$$

$$\hat{y}_{\bar{x}} \sim N\left[y_{\bar{x}}, Var(\hat{y}_{\bar{x}})\right]$$

$$T = \frac{\hat{y}_{\bar{x}} - y_{\bar{x}}}{se(\hat{y}_{\bar{x}})} \sim t(3)$$

$y_{\bar{x}}$ 之95%預測區間: $\left[\hat{y}_{\bar{x}} - se(\hat{y}_{\bar{x}})\cdot t_{0.025}(3),\ \hat{y}_{\bar{x}} + se(\hat{y}_{\bar{x}})\cdot t_{0.025}(3)\right]$

$$= \left[2 - \sqrt{1.44} \times 3.182,\ 2 + \sqrt{1.44} \times 3.182\right]$$

$$= \left[-1.8184,\ 5.8184\right]$$

The width of interval (c) : $se(\hat{y}_0)\cdot t_{0.025}(3) = 1.3856 \times 3.182 = 4.4090$

The width of interval (e) : $se(\hat{y}_{\bar{x}})\cdot t_{0.025}(3) = \sqrt{1.44} \times 3.182 = 3.8184$

$4.4090 > 3.8184$ $\therefore$ $x = \bar{x}$ 的區間寬度較小

4.25

(a)

```
Call:
lm(formula = log(price) ~ sqft, data = collegetown)

Residuals:
     Min       1Q   Median       3Q      Max
-1.32528 -0.19199  0.00122  0.21678  0.86609

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 4.393866   0.043288  101.50   <2e-16 ***
sqft        0.036044   0.001486   24.26   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.34 on 498 degrees of freedom
Multiple R-squared:  0.5417,	Adjusted R-squared:  0.5408
F-statistic: 588.7 on 1 and 498 DF,  p-value: < 2.2e-16

> slope_a<-coef(log_linear)[[2]]*avg_PRICE
> elas_a<-coef(log_linear)[[2]]*avg_SQFT
> cat("a小題斜率為:",slope_a,"\n")
a小題斜率為: 9.019661
> cat("a小題彈性為:",elas_a,"\n")
a小題彈性為: 0.9833701
```

$$\ln(PRICE) = \beta_1 + \beta_2 SQFT + e$$

$$\widehat{\ln(PRICE)} = \hat{\beta}_1 + \hat{\beta}_2 SQFT$$

$$= 4.393866 + 0.036044\, SQFT$$

$\hat{\beta}_1$ 表示在 SQFT=0 時,預期房價為 $e^{4.393866} = 84.7636$

$\hat{\beta}_2$ 表示每增加1單位SQFT,預期房價會上升 $0.036044 \times 100\% = 3.6044\%$

$$slope = \frac{dy}{dx} = \hat{\beta}_2 \times \overline{PRICE} = 9.019661$$

$$elasticity = \frac{dy}{dx}\cdot\frac{x}{y} = \hat{\beta}_2 \times \overline{SQFT} = 0.9833701$$

(b)

```
Call:
lm(formula = log(price) ~ log(sqft), data = collegetown)

Residuals:
     Min       1Q   Median       3Q      Max 
-1.36864 -0.20849  0.00487  0.23204  1.21099 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept)  2.04965    0.15797   12.97   <2e-16 ***
log(sqft)    1.02483    0.04839   21.18   <2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.3643 on 498 degrees of freedom
Multiple R-squared:  0.4738,	Adjusted R-squared:  0.4728 
F-statistic: 448.5 on 1 and 498 DF,  p-value: < 2.2e-16

> slope_b<-coef(log_log)[[2]]*avg_PRICE/avg_SQFT
> elas_b<-coef(log_log)[[2]]
> cat("b小題斜率為:",slope_b,"\n")
b小題斜率為: 9.399923
> cat("b小題彈性為:",elas_b,"\n")
b小題彈性為: 1.024828
```

$$\ln(PRICE) = \alpha_1 + \alpha_2 \ln(SQFT) + e$$

$$\widehat{\ln(PRICE)} = \hat{\alpha}_1 + \hat{\alpha}_2 \ln(SQFT)$$

$$= 2.04965 + 1.02483$$

$\hat{\alpha}_1$ 表示在 $SQFT = 1$ 時，預期房價為 $e^{2.04965} = 7.7652$

$\hat{\alpha}_2$ 表示每增加 1% SQFT，預期房價會上升 1.02483%

$$slope = \frac{dy}{dx} = \hat{\alpha}_2 \cdot \frac{\overline{PRICE}}{\overline{SQFT}} = 9.399923$$

$$elasticity = \frac{dy}{dx} \cdot \frac{x}{y} = \hat{\alpha}_2 = 1.02483$$

(c)

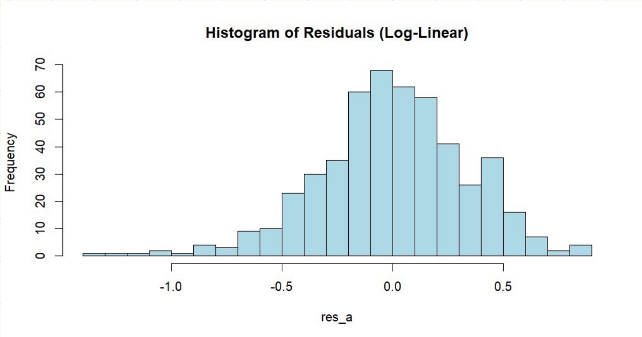
```
> r2_linear <- summary(linear_linear)$r.squared
> #為了將應變數變為一致，先得到ln(price)的預測值，再取exp()轉回原始價格單位
> pred_price_loglog<-exp(predict(log_log))
> # 計算實際價格與預測價格相關係數的平方
> r2_gen_loglog <- cor(collegetown$price, pred_price_loglog)^2
> cat("線性模型之R平方為:",r2_linear,"\n")
線性模型之R平方為: 0.6413167
> cat("對數-對數之廣義R平方為:",r2_gen_loglog,"\n")
對數-對數之廣義R平方為: 0.6445084
```

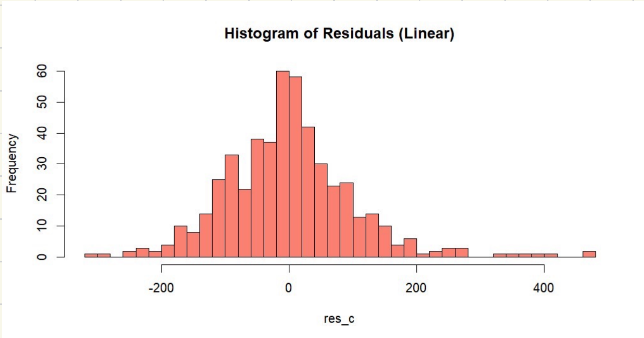
model (B) generalized R^2 : 0.6445084

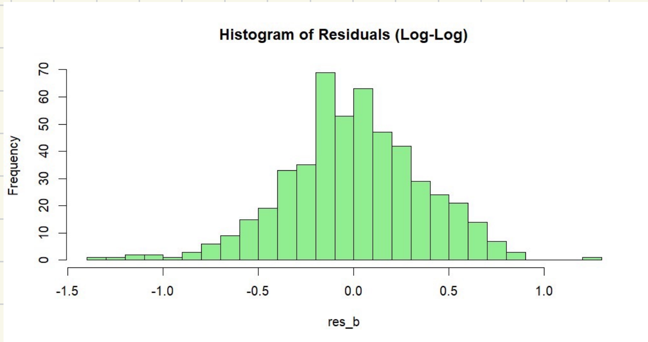
model (C) R^2 : 0.6413167

說明 log-log model 對於房價變動的解釋能力略優於 linear model #

(d)







```
> jb_a <-jarque.bera.test(res_a)
> jb_b <-jarque.bera.test(res_b)
> jb_c <-jarque.bera.test(res_c)
> print(jb_a)

        Jarque Bera Test

data:  res_a
X-squared = 26.679, df = 2, p-value = 1.61e-06

> print(jb_b)

        Jarque Bera Test

data:  res_b
X-squared = 14.279, df = 2, p-value = 0.0007933

> print(jb_c)

        Jarque Bera Test

data:  res_c
X-squared = 221.11, df = 2, p-value < 2.2e-16
```

H_0: model_a 為常態分配 vs H_1: model_a 非為常態分配

$\because$ P-value $= 1.61\times10^{-6} < 0.05$ $\therefore$ reject H_0 at $\alpha=0.05$

表示有足夠證據顯示 model_a 之分配非為常態分配

H_0: model_b 為常態分配 vs H_1: model_b 非為常態分配

$\because$ P-value $= 0.0007933 < 0.05$ $\therefore$ reject H_0 at $\alpha=0.05$

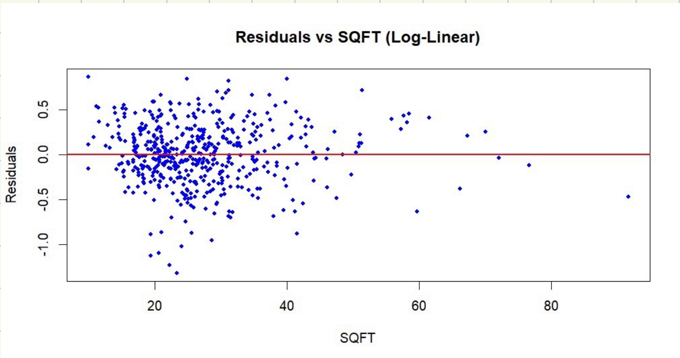
表示有足夠證據顯示 model_b 之分配非為常態分配

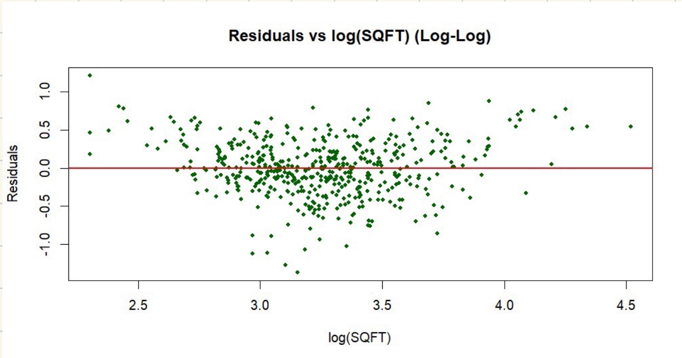
H_0: model_C 為常態分配 vs H_1: model_c 非為常態分配

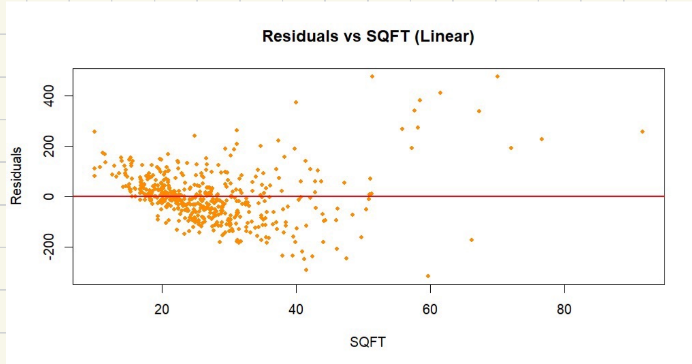
$\because$ P-value $= 2.2\times10^{-16} < 0.05$ $\therefore$ reject H_0 at $\alpha=0.05$

表示有足夠證據顯示 model_c 之分配非為常態分配

(e)







在 linear model 的殘差圖中可觀察 隨著SQFT增加，殘差的散佈範圍變得愈大，存在異質變異性，代表 linear model 在預測大坪數房屋時，誤差會顯著增加。

log-log 與 log-linear model 不會隨著SQFT增加，殘差的散佈範圍更大，較符合同質變異性。

(f)

```
> #f小題
> new_house<-data.frame(sqft = 27)
> pred_log_a<-predict(log_linear,newdata=new_house)
> #取指數轉回原始房價
> price_a<-exp(pred_log_a)
>
> pred_log_b<-predict(log_log, newdata=new_house)
> price_b<-exp(pred_log_b)
>
> pred_c<-predict(linear_linear,newdata=new_house)
> price_c<-pred_c
> cat("Log-Linear 預測房價:", price_a, "\n")
Log-Linear 預測房價: 214.2336
> cat("Log-Log     預測房價:", price_b, "\n")
Log-Log     預測房價: 227.5386
> cat("Linear      預測房價:", price_c, "\n")
Linear      預測房價: 246.4557
```

(g)

```
> PI_a<-predict(log_linear, new_house, interval = "prediction", level = 0.95)
> target_PI_a<-exp(PI_a)
>
> PI_b<-predict(log_log, new_house, interval = "prediction", level = 0.95)
> target_PI_b<-exp(PI_b)
>
> target_PI_c<-predict(linear_linear, new_house, interval = "prediction", level=0.95)
>
> cat("model a之預測區間下界為:",target_PI_a[2],"上界為:",target_PI_a[3],"\n")
model a之預測區間下界為: 109.7685 上界為: 418.1167
> cat("model b之預測區間下界為:",target_PI_b[2],"上界為:",target_PI_b[3],"\n")
model b之預測區間下界為: 111.1406 上界為: 465.8406
> cat("model c之預測區間下界為:",target_PI_c[2],"上界為:",target_PI_c[3],"\n")
model c之預測區間下界為: 44.27727 上界為: 448.6341
```

(h)

我會選擇Log-Log model，此模型不僅在Jarque-Bera檢定與廣義R^2上表現優異，且成功消除線性模型中明顯的異質變異性，使預測區間與推論更具可靠性。